On my honor, I have neither given nor received any unauthorized and/or inappropriate assistance for all sessions of this exam. The work done on this exam is totally my own. I understand that by the school code, violation of these principles will lead to a zero grade and is subject to harsch discipline issues."

1.

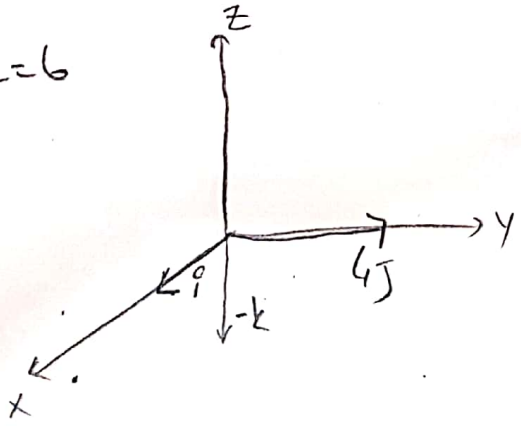
$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & k & 0 & 1 \end{bmatrix} \xrightarrow{R_2+R_3 \to R_3} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & k & 0 & 1 \end{bmatrix}$$

2)

a) There are 2 valuable as $\overset{2 \text{ axis}}{x \text{ and } y}$. So it has to be 2 dimensional at least.

b) If I propose $x^2+y^2=z$, it states a parabolic function and in a 3 dimensional space. it has a volume.

c) $x+4y-z=6$



Yes, these vectors form a vector space.

3 dimensions inside 3 dimensional space.

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3) Assume $n=2$ 2×2 unit vector $u=\begin{bmatrix}1\\0\end{bmatrix}$

$$H=\begin{bmatrix}1&0\\0&1\end{bmatrix}-2\cdot\begin{bmatrix}1&0\\0&1\end{bmatrix}\cdot\begin{bmatrix}1&0\\0&1\end{bmatrix}=\begin{bmatrix}1&0\\0&1\end{bmatrix}-\begin{bmatrix}2&0\\0&2\end{bmatrix}=\begin{bmatrix}-1&0\\0&-1\end{bmatrix}$$

a) $H^2=\begin{bmatrix}-1&0\\0&-1\end{bmatrix}\begin{bmatrix}-1&0\\0&-1\end{bmatrix}=\begin{bmatrix}1&0\\0&1\end{bmatrix}=I$

b) We should separate to its columns to create vectors. Features of orthonormal matrix :

$1\longrightarrow v_1\cdot v_2=0 \longrightarrow \begin{pmatrix}-1\\0\end{pmatrix}\cdot\begin{pmatrix}0\\-1\end{pmatrix}=\begin{pmatrix}0\\0\end{pmatrix}=0$ $\boxed{v_1\perp v_2}$

$2\longrightarrow |v_1|=1,\ |v_2|=1$ $\quad \left|\begin{matrix}-1\\0\end{matrix}\right|=\sqrt{1+0}=1 \quad \left|\begin{matrix}0\\-1\end{matrix}\right|=\sqrt{0+1}=1$

c) $\begin{bmatrix}-1-\lambda&0\\0&-1-\lambda\end{bmatrix}\cdot\begin{bmatrix}1&0\\0&1\end{bmatrix}=0 \quad \begin{bmatrix}-1-\lambda&0\\0&-1-\lambda\end{bmatrix}=\begin{bmatrix}0&0\\0&0\end{bmatrix}$

$-1-\lambda=0$

$\boxed{\lambda=-1}$

4) a) $\begin{bmatrix}v_{k+1}\\w_{k+1}\end{bmatrix}=\begin{bmatrix}-2&1\\1&-2\end{bmatrix}\cdot\begin{bmatrix}v_k\\w_k\end{bmatrix}$

b) $v_{k+1}=-2v_k+w_k$ $\boxed{k\to0}$

$V_1=-2\cdot v_0+w_0 \Rightarrow v_1=-2(10)+(40)\Rightarrow 20$ liters of mixture

It was 10 liters of liquid. Then we add 10 liters of water in it. So percentages of liquid and water are same and it %50.

c) For steady state, first of all we should find the eigenvectors.

$|A-\lambda I|=0 \quad \begin{vmatrix}-2-\lambda&1\\1&-2-\lambda\end{vmatrix}=0 \quad (-2-\lambda)^2-1=4+4\lambda+\lambda^2-1$

$=\lambda^2+4\lambda+3=0$

$(\lambda+3)(\lambda+1)=0$

$\lambda_1=-3 \quad \lambda_2=-1$

for $\lambda_1=-3$

$$\begin{bmatrix}-2+3&1\\1&-2+3\end{bmatrix}=\begin{bmatrix}1&1\\1&1\end{bmatrix}\cdot x_1=0 \qquad \boxed{x_1=\begin{bmatrix}a\\-a\end{bmatrix}=a\begin{bmatrix}1\\-1\end{bmatrix}}$$

for $\lambda_2=-1$

$$\begin{pmatrix}-2+1&1\\1&-2+1\end{pmatrix}=\begin{bmatrix}-1&1\\1&-1\end{bmatrix}\cdot x_2=0 \qquad x_2=\begin{bmatrix}b\\b\end{bmatrix}=b\begin{bmatrix}1\\1\end{bmatrix}$$

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$$\begin{bmatrix} v_k \\ w_k \end{bmatrix} = c_1 \lambda_1 x_1 + c_2 \lambda_2 x_2 = c_1 \cdot \begin{bmatrix} 1 \\ -1 \end{bmatrix} \cdot (-3)^k + c_2 \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} \cdot (-1)^k$$

for $k = 0$ $\quad \begin{bmatrix} v_0 \\ w_0 \end{bmatrix} = \begin{bmatrix} 10 \\ 40 \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ -1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$c_1 + c_2 = 10$

$c_2 - c_1 = 40$

$2c_2 = 50$

$c_2 = 25$

$c_1 = -15$

$$\begin{bmatrix} v_k \\ w_k \end{bmatrix} = -15 \begin{bmatrix} 1 \\ -1 \end{bmatrix} (-3)^k + 25 \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} (-1)^k$$

if $k \to \infty$ $\quad \begin{bmatrix} v_\infty \\ w_\infty \end{bmatrix} = \begin{bmatrix} -15 \cdot (-3)^k \\ 15 \cdot (-3)^k \end{bmatrix} + \begin{bmatrix} 25(-1)^k \\ 25(-1)^k \end{bmatrix}$

I did all the process till this point. I couldn't do the final step.

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